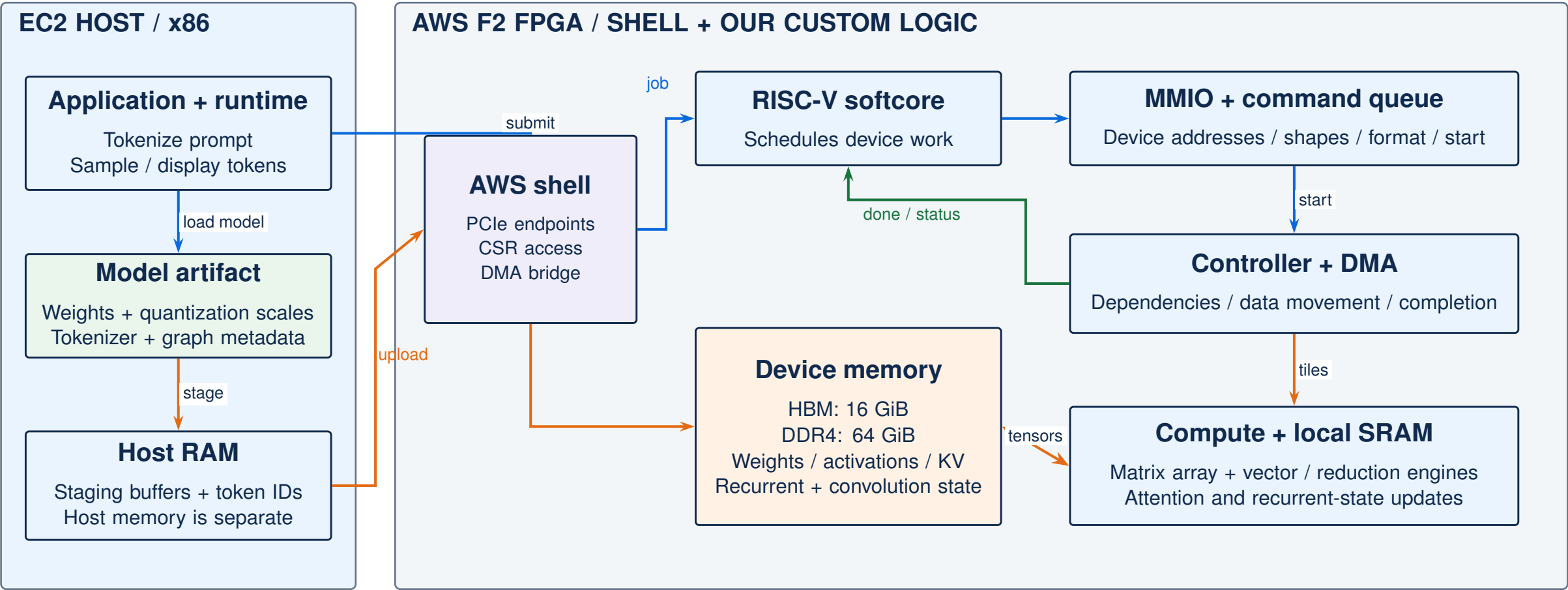


One system, two processors, one accelerator

The CPU executes instructions. The accelerator receives commands and tensors.



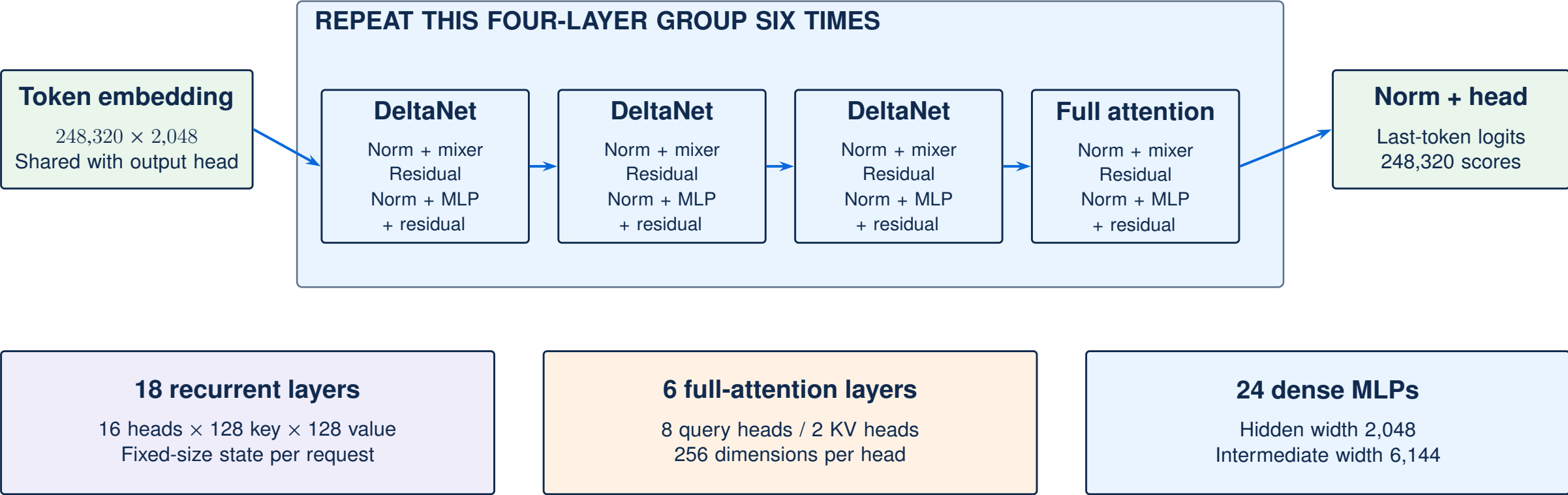
Blue: control **Orange: tensor data** **Green: completion** RISC-V receives completions; the host receives output tokens.

Takeaway. Keep weights and sequence state on the device. Use the host link for setup, requests, and results.

Sources and evidence: [1-4].

Qwen3.5-2B is a hybrid model

An accelerator for this model must handle dense matrices, attention, and recurrent state.



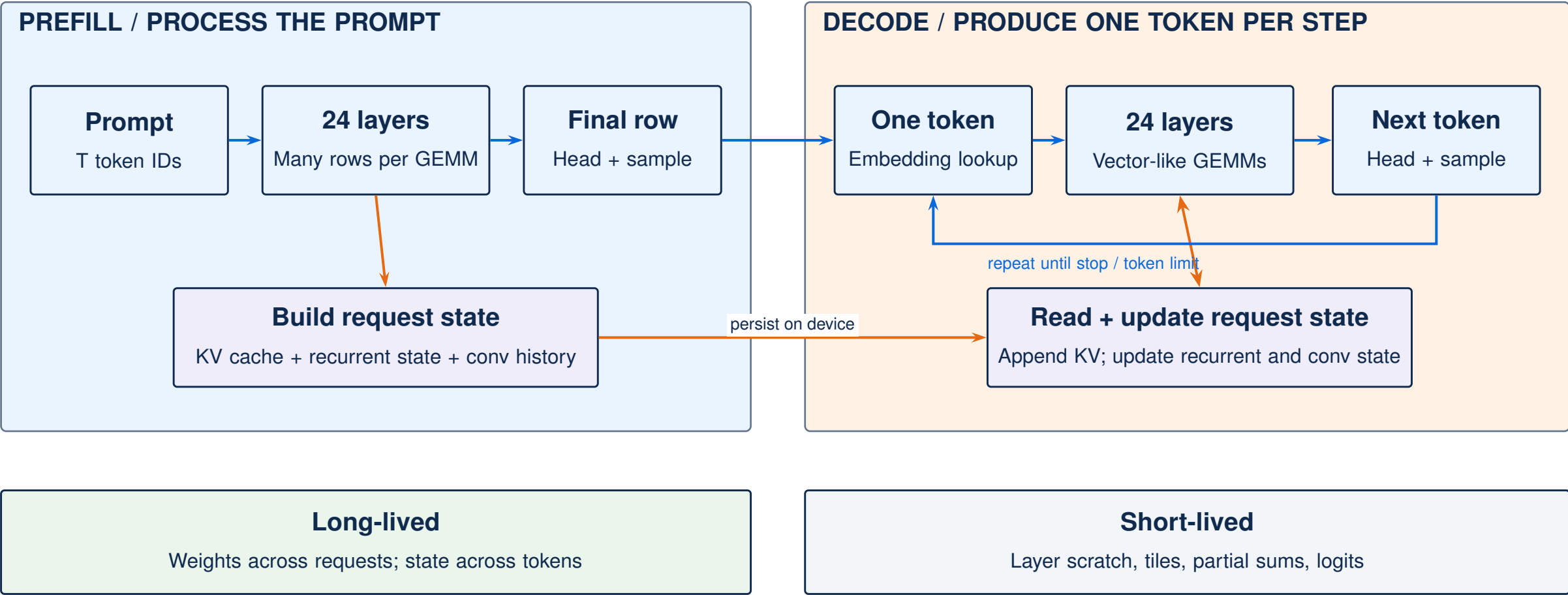
Text model: 1.882 billion unique parameters. The full 2.274 billion-parameter checkpoint also contains vision and MTP parameters. Text inference is the current scope.

Takeaway. A single MLP is a useful test block. It is only one part of the complete 24-layer inference path.

Sources and evidence: [5-7].

The inference loop has two different workloads

Prefill creates sequence state. Decode reuses it while generating new tokens.

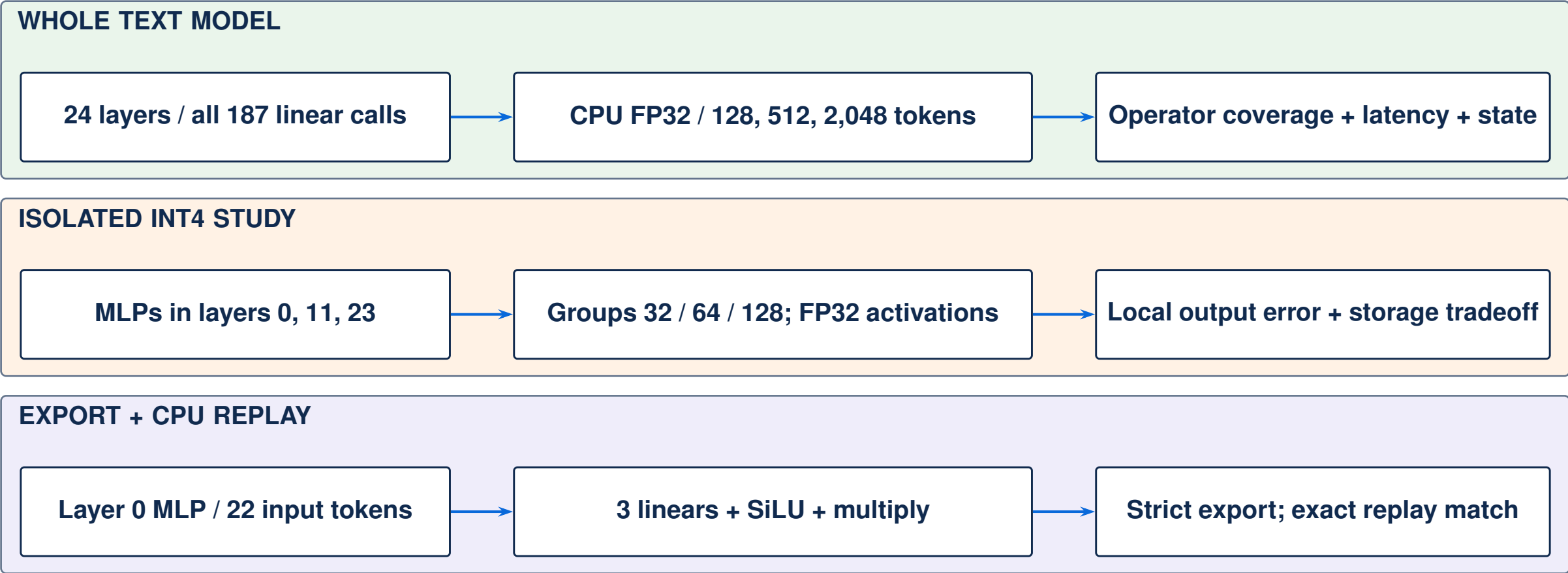


Takeaway. Schedule and benchmark prefill and decode separately. They have different reuse, latency, and memory behavior.

Sources and evidence: [6, 8].

What the experiments actually established

All three evidence sets are useful, but they answer different questions.

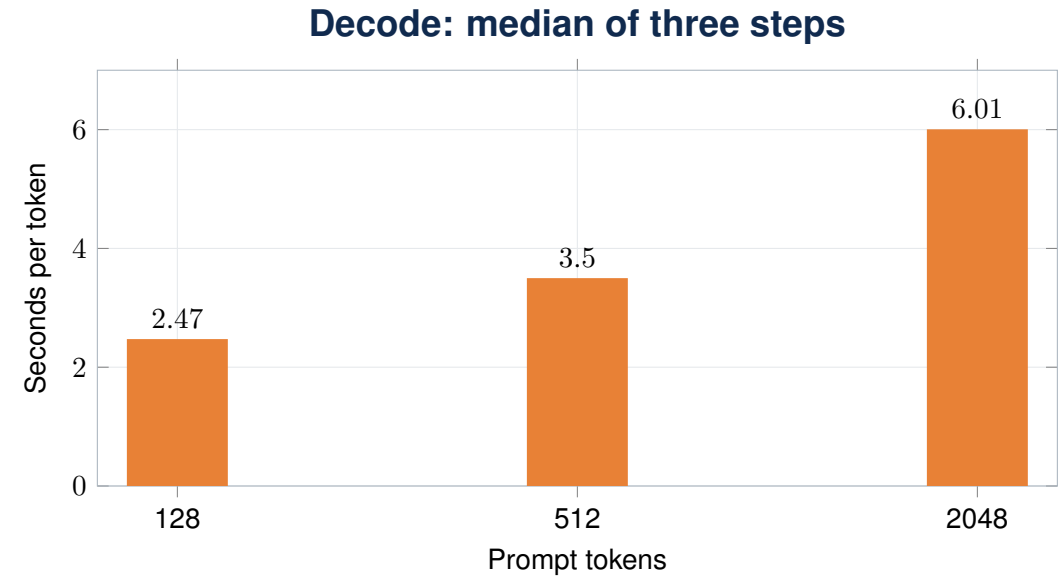
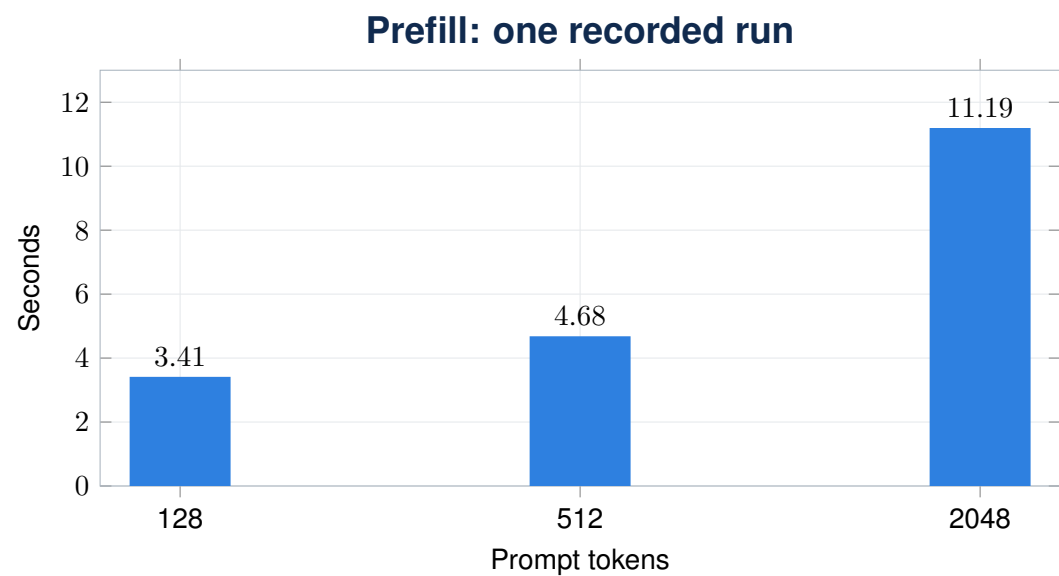


Takeaway. Full-model coverage is measured. Full-model INT4 quality, FPGA speed, and RISC-V execution remain unmeasured.

Sources and evidence: [\[8-10\]](#).

Measured latency is a baseline, not a hardware forecast

The traces cover the full text network. These timings include CPU-specific fallback behavior.



Measured environment

Apple M5 Pro / CPU FP32 / 6 threads / batch 1
Eager attention; optional fused DeltaNet kernels absent

Read the result correctly

Not an FPGA estimate or a controlled scaling study
Load, tokenization, sampling, and trace I/O excluded

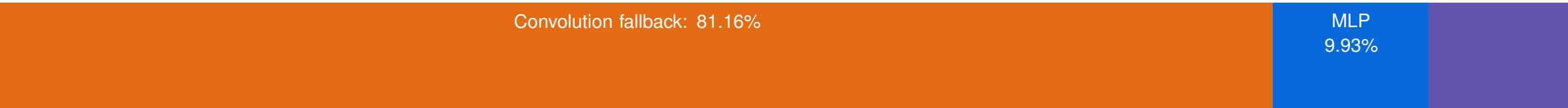
Takeaway. Use the traces to validate graph coverage and locate software bottlenecks. Re-measure each hardware mapping.

Sources and evidence: [8, 11].

Time spent on a CPU is different from arithmetic work

A slow fallback kernel must not dictate how much silicon each operation receives.

Decode at a 2,048-token prompt: instrumented wall-time categories



Other time: DeltaNet projections 4.60%; head 2.13%; attention projections 0.96%; remaining work 1.22%.

Dense linear work per decoded token: 1.881 billion MACs



The orange slice is full-attention projections. Counts exclude attention QK/AV, recurrence, convolution, and nonlinear operations.

Two measurements, two decisions

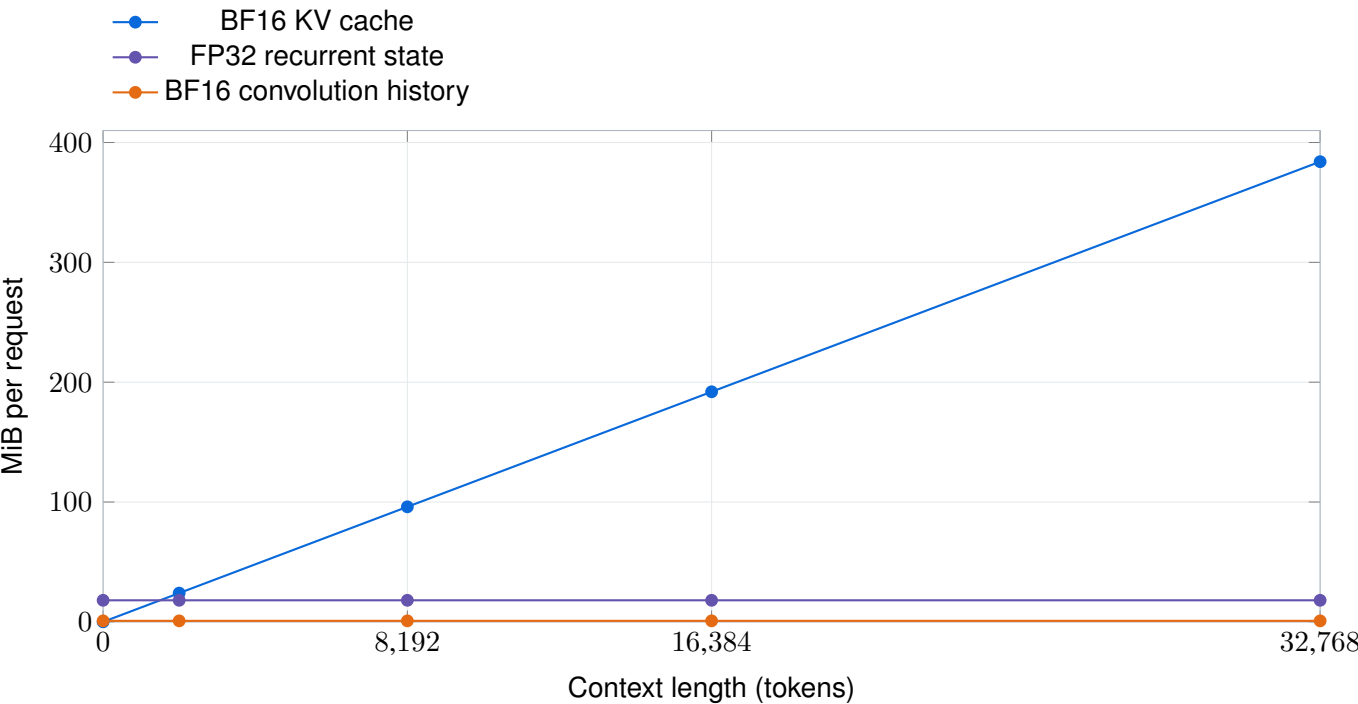
Fix CPU convolution for software speed; include the output head when sizing matrix hardware.

Takeaway. The output head is 27% of dense MACs but only 2% of measured CPU time. Neither percentage alone sizes the accelerator.

Sources and evidence: [6, 8].

Three kinds of state, three memory behaviors

The capacity plot assumes BF16 KV and convolution history, with FP32 recurrent state.



8K planning case: 114.84 MiB per request.
The experiments used FP32 caches: 22.69 / 31.69 / 67.69 MiB at 128 / 512 / 2,048 tokens.

KV grows with context

$$6 \cdot 2 \cdot 2 \cdot 256 \cdot T \cdot 2 \text{ bytes} = 12,288T \text{ bytes at BF16}$$

Recurrent state stays fixed

$$18 \cdot 16 \cdot 128 \cdot 128 \cdot 4 \text{ bytes} = 18 \text{ MiB at FP32}$$

Convolution keeps four slots

$$18 \cdot 6,144 \cdot 4 \cdot 2 \text{ bytes} = 0.84375 \text{ MiB at BF16}$$

Takeaway. KV is only part of the persistent state. Admission control and context switching must account for all three stores.

Sources and evidence: [\[6-8\]](#).

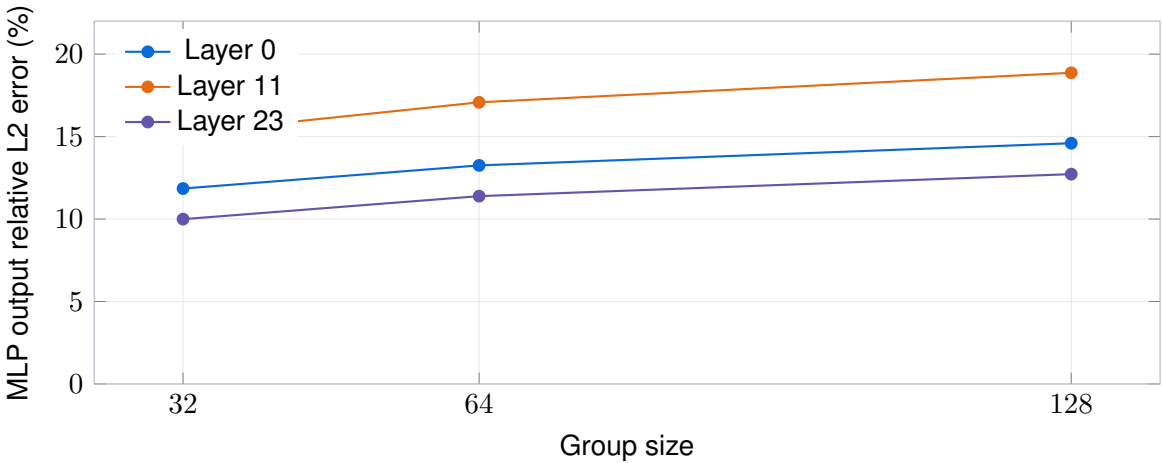
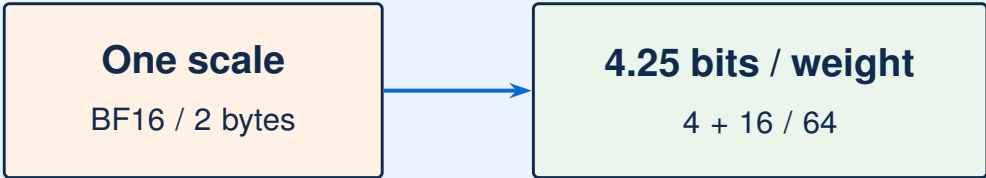
INT4 saves bandwidth; scale groups cost bytes

The recorded study quantized three isolated MLPs, then replayed them with FP32 activations.

ONE GROUP OF 64 WEIGHTS



64 four-bit codes = 32 bytes



Smaller groups
More scale traffic
Lower error in these nine local tests

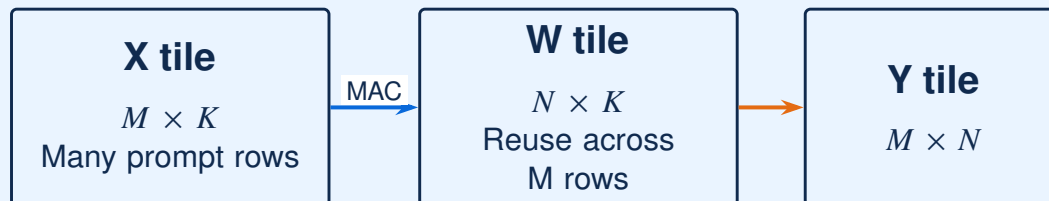
What is still open
Whole-model quality / FP4 comparison
Activation precision / accumulator precision

Takeaway. Group 64 is a planning baseline, not a quality sign-off. Relative MLP output error is not model accuracy loss.

Sources and evidence: [9, 12-15].

The same array needs two scheduling strategies

Tile reuse favors prefill. Weight delivery often dominates single-request decode.

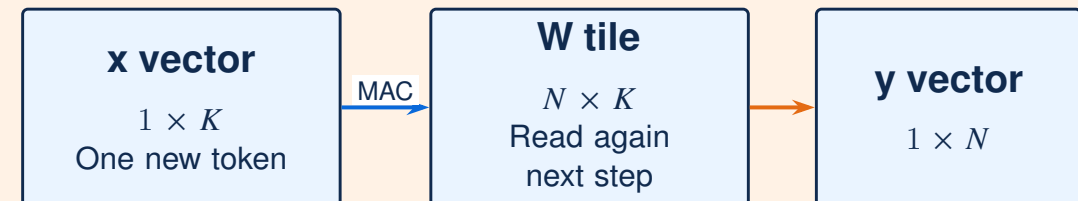


Reuse opportunity

For INT4 weights, weight-only intensity is approximately $2M$ MAC/byte before scales.

Main tradeoff

Larger tiles reuse weights but need more activation and accumulator SRAM.



Bandwidth pressure

At batch 1, weight-only intensity is approximately 2 MAC/byte before scales.

Main tradeoff

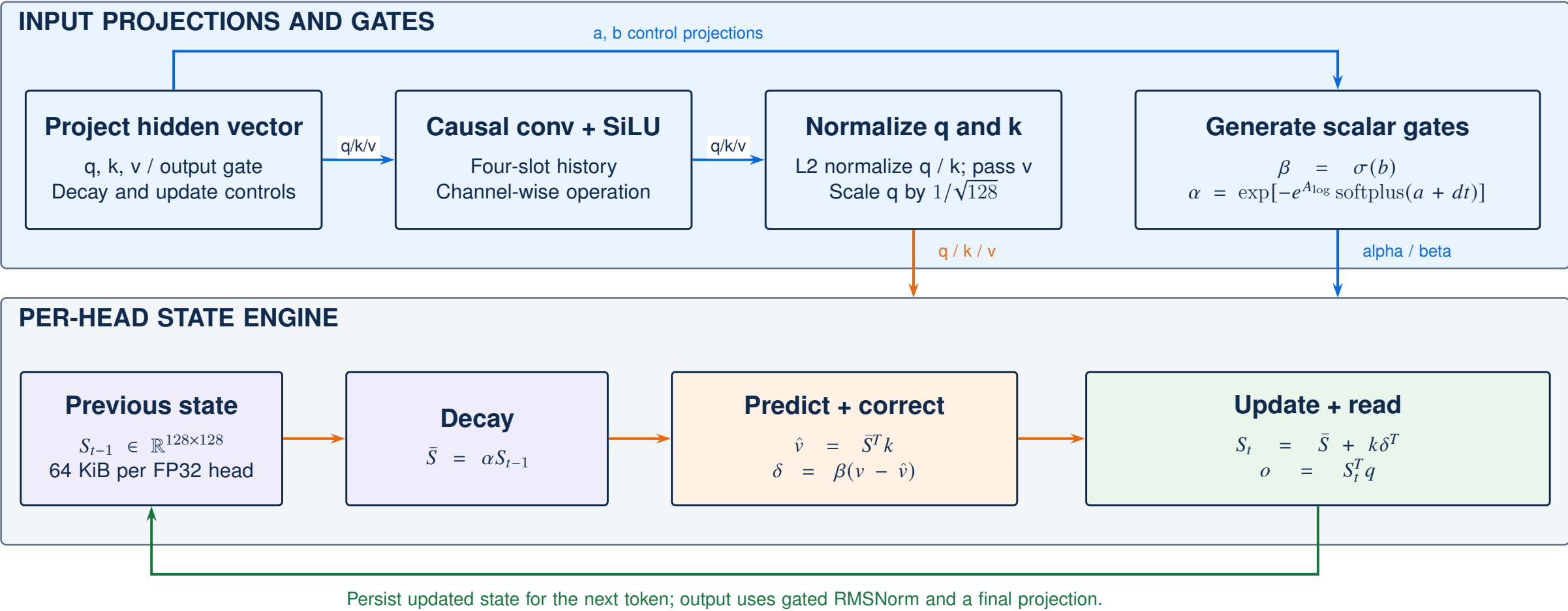
Batching amortizes weight reads but adds request state and queueing latency.

Takeaway. Design for useful utilization in both modes. A large peak-MAC number cannot overcome a starved weight path.

Sources and evidence: [7, 8].

DeltaNet needs a state engine as well as a matrix array

Per-head equations shown in key-by-value state orientation; 16 heads in each of 18 layers.

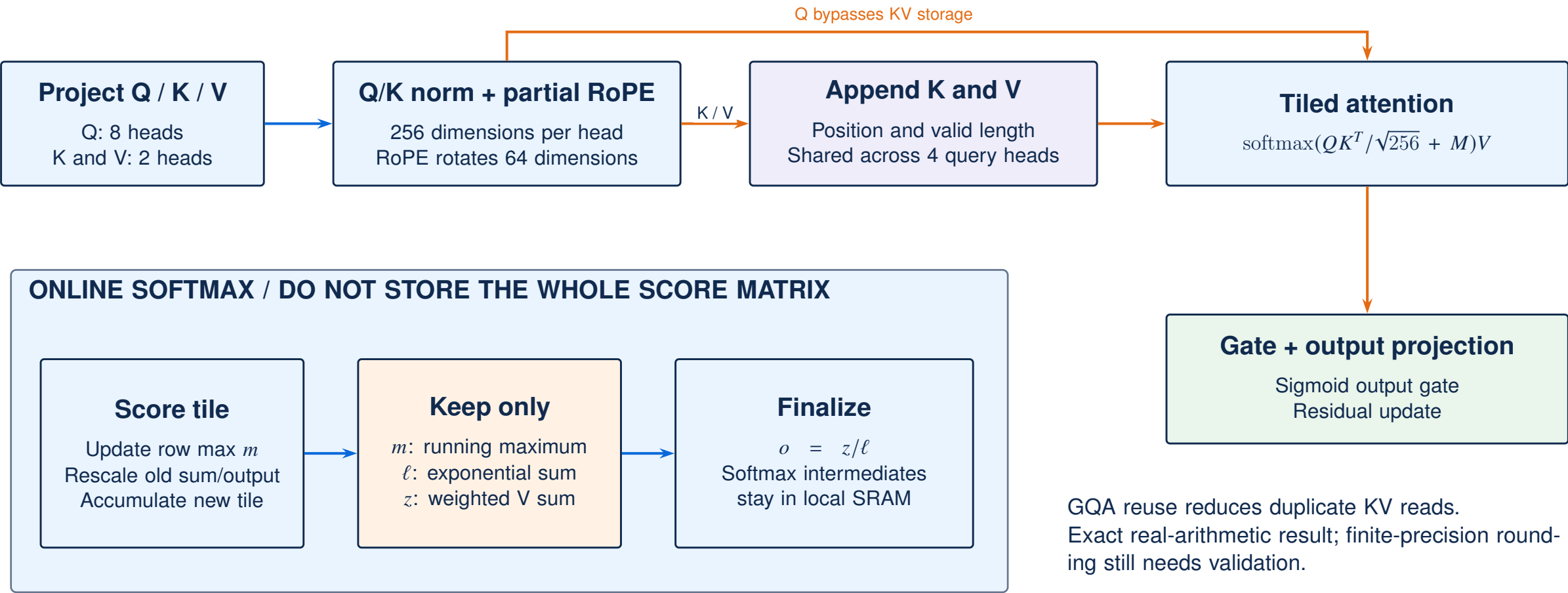


Takeaway. A token changes recurrent state irreversibly without a saved checkpoint. Reset, replay, and context switching need explicit rules.

Sources and evidence: [6, 16].

Attention is a streaming reduction problem

Grouped-query attention and tiled softmax reduce traffic without changing the model graph.

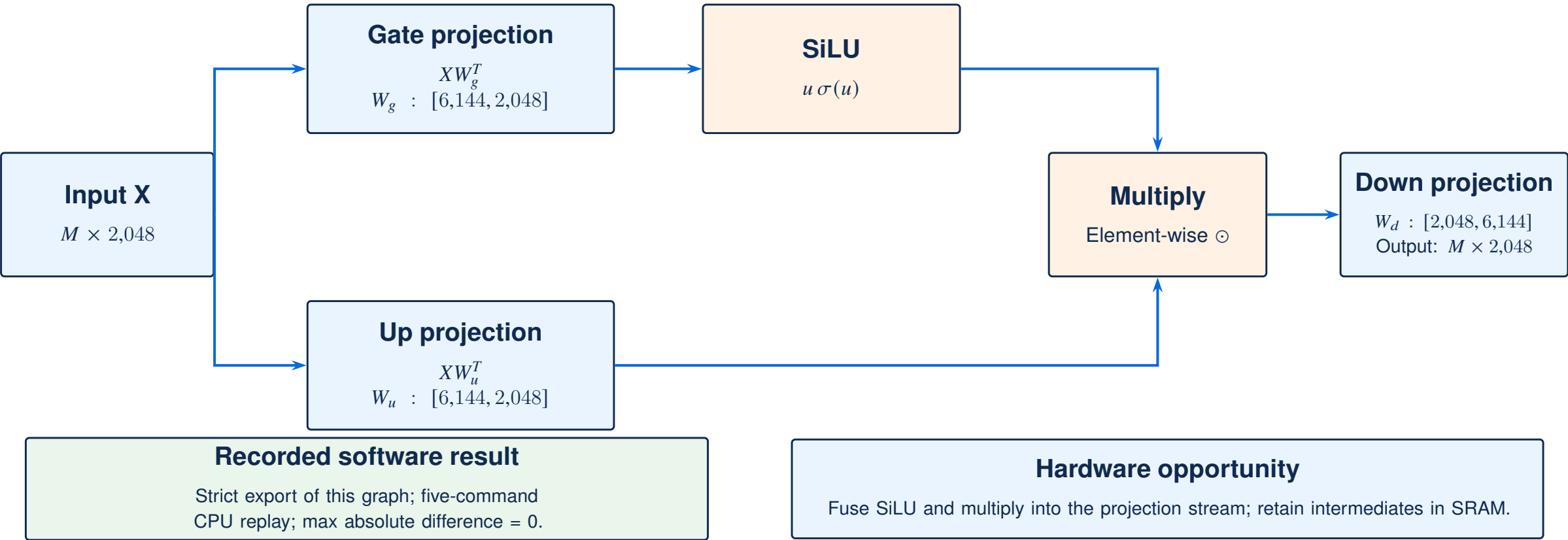


Takeaway. Avoid materializing a full T-by-T score matrix. Validate masking, positions, Q/K normalization, and output gating.

Sources and evidence: [6, 17, 18].

The MLP experiment demonstrates one reusable mapping

Three matrix operations and two element-wise operations become an explicit command sequence.



Takeaway. Extend the same correctness-first mapping to attention, recurrence, embeddings, norms, and the output head.

Sources and evidence: [9, 10, 19].

Memory is managed explicitly, not as one shared RAM

Host pointers, FPGA device addresses, and local SRAM indices belong to different address spaces.



LIFETIME DETERMINES ALLOCATION

- Model lifetime** Weights, scales, layer descriptors.
- Request lifetime** KV, recurrent state, conv history, position.
- Layer lifetime** Activations and scratch workspace.
- Tile lifetime** Inputs, dequantized values, accumulators.

BANK MAPPING DETERMINES THROUGHPUT



Proposed partition, not fixed physical addresses.
Stripe for parallel ports; measure bank conflicts and burst efficiency.

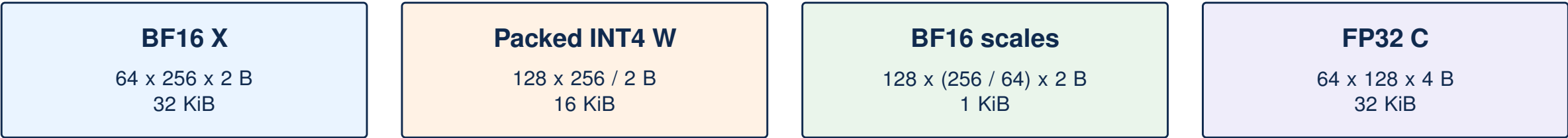
Takeaway. Allocate by lifetime, map by bank, and transfer by ownership. A CPU pointer is not automatically a DMA address.

Sources and evidence: [2, 7, 20].

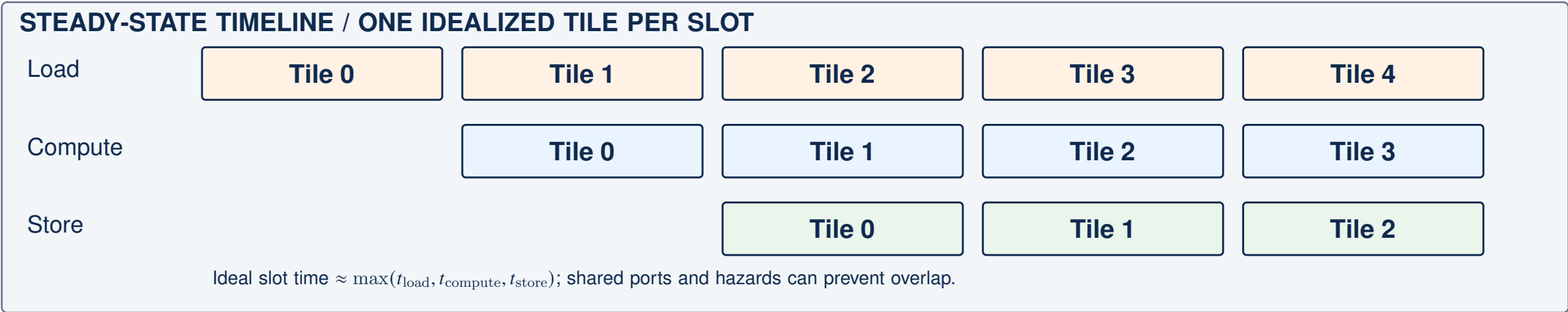
Local SRAM buys reuse and overlap

Buffer capacity must include packed weights, scales, activations, and wider partial sums.

Illustrative GEMM tile: $M = 64, N = 128, K = 256$



Double-buffer inputs and weights: $2(32 + 16 + 1) + 32 = 130$ KiB before alignment, queues, and other engines.

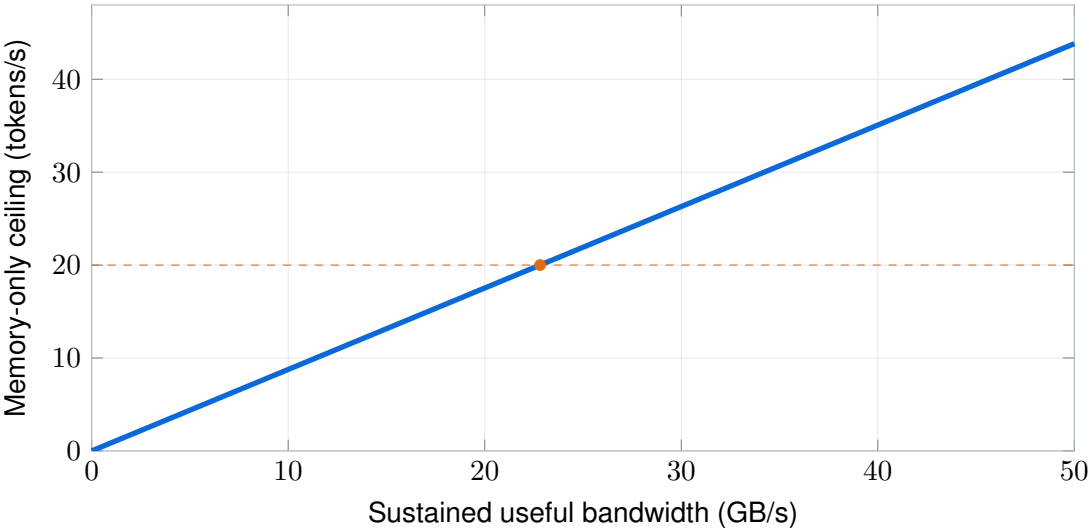
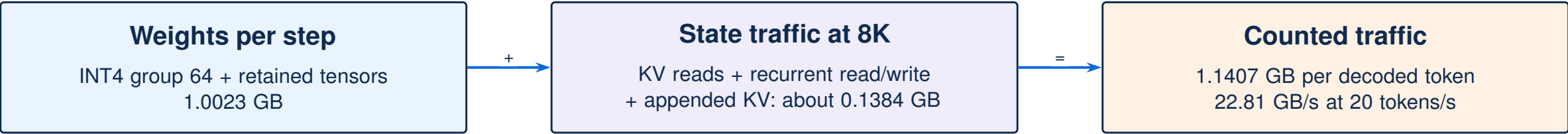


Takeaway. Double buffering helps only when memory, compute, and writeback can make progress at the same time.

Sources and evidence: [7, 20].

Capacity is not enough: bytes must arrive on time

This lower-bound traffic model counts weights, attention KV, and recurrent-state traffic.



Compute check at the same scenario

Counted work: 2.097 GMAC/token

20 tokens/s: 41.94 GMAC/s

250 MHz, 60% utilization: about 280 MAC lanes

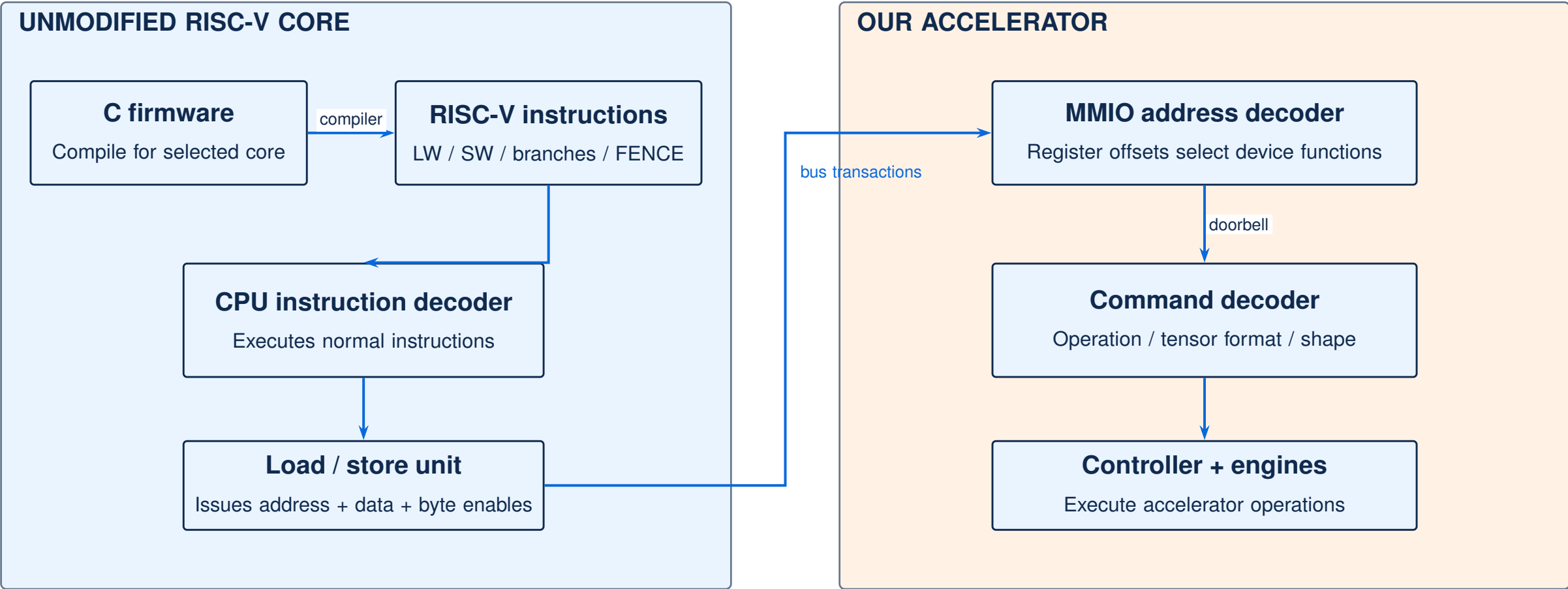
20 tokens/s is an illustrative design point.
Traffic omits scratch spills, padding, and protocol overhead; assumes GQA reuse and one weight pass.

Takeaway. Size memory and compute together. Benchmark sustained useful bytes, not only bus width or advertised bandwidth.

Sources and evidence: [7, 8].

The accelerator does not decode RISC-V instructions

MMIO uses ordinary loads and stores. Custom instructions and the vector extension are not required.

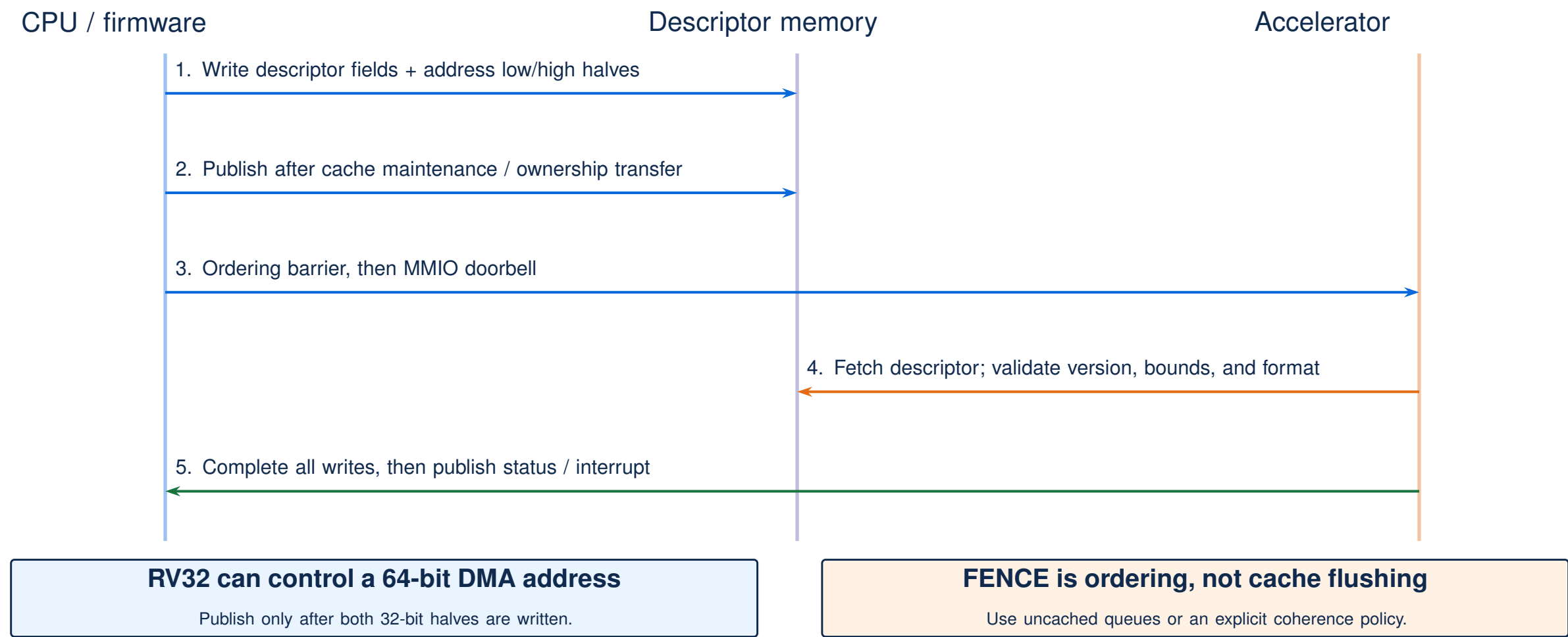


Takeaway. The CPU decodes its ISA; our hardware decodes its register and command interface. Those are separate contracts.

Sources and evidence: [3, 21-24].

A command needs an ownership protocol

Example single-producer sequence; actual register offsets and queue ABI remain to be frozen.

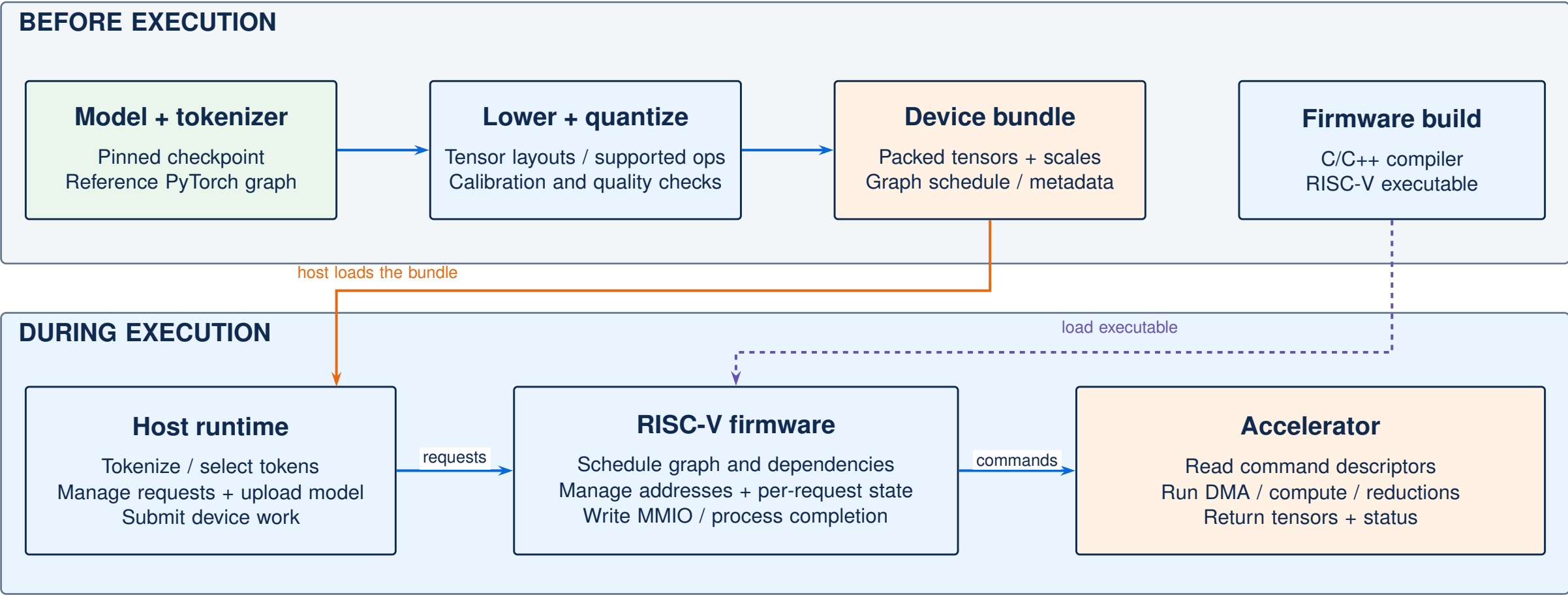


Takeaway. Use sequence numbers and explicit address domains. Completion must mean results are visible before a buffer is reused.

Sources and evidence: [3, 21, 23].

From PyTorch to hardware: two compilation paths

The model becomes tensors and an execution schedule. Firmware becomes a RISC-V program.

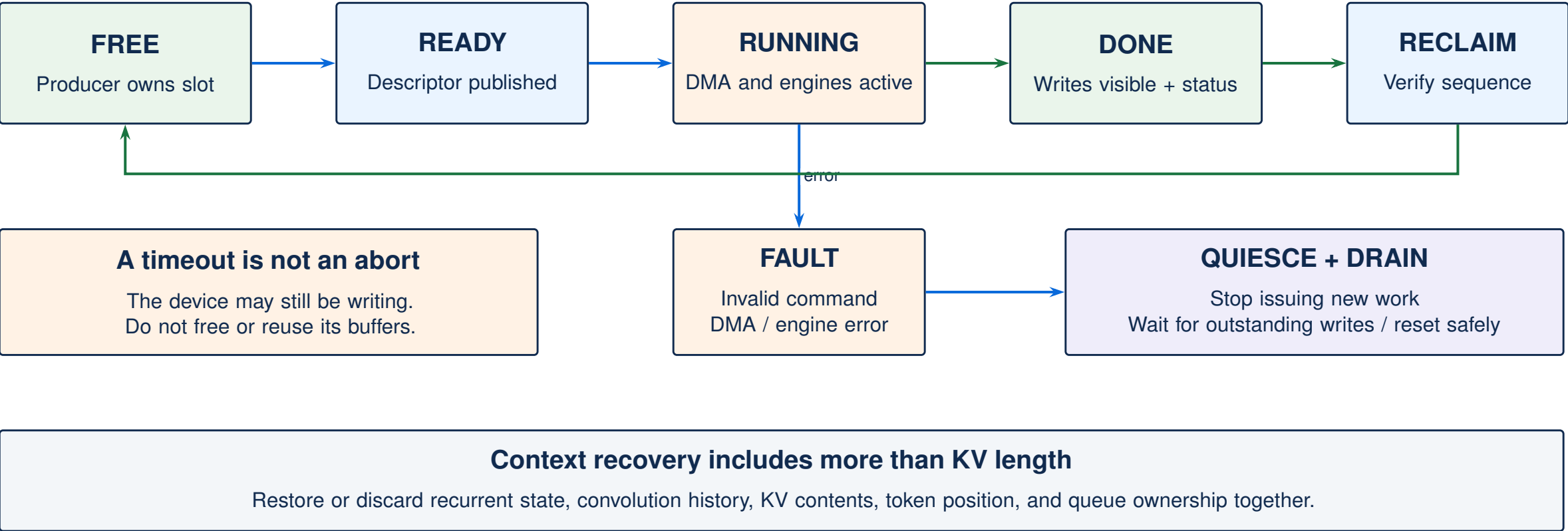


Takeaway. A llama.cpp backend or exported-graph runtime must implement the full operator and state contract. C alone does not supply it.

Sources and evidence: [4, 10, 19, 25].

Control logic must handle incomplete work safely

Submission, completion, cancellation, reset, and replay are part of the hardware contract.

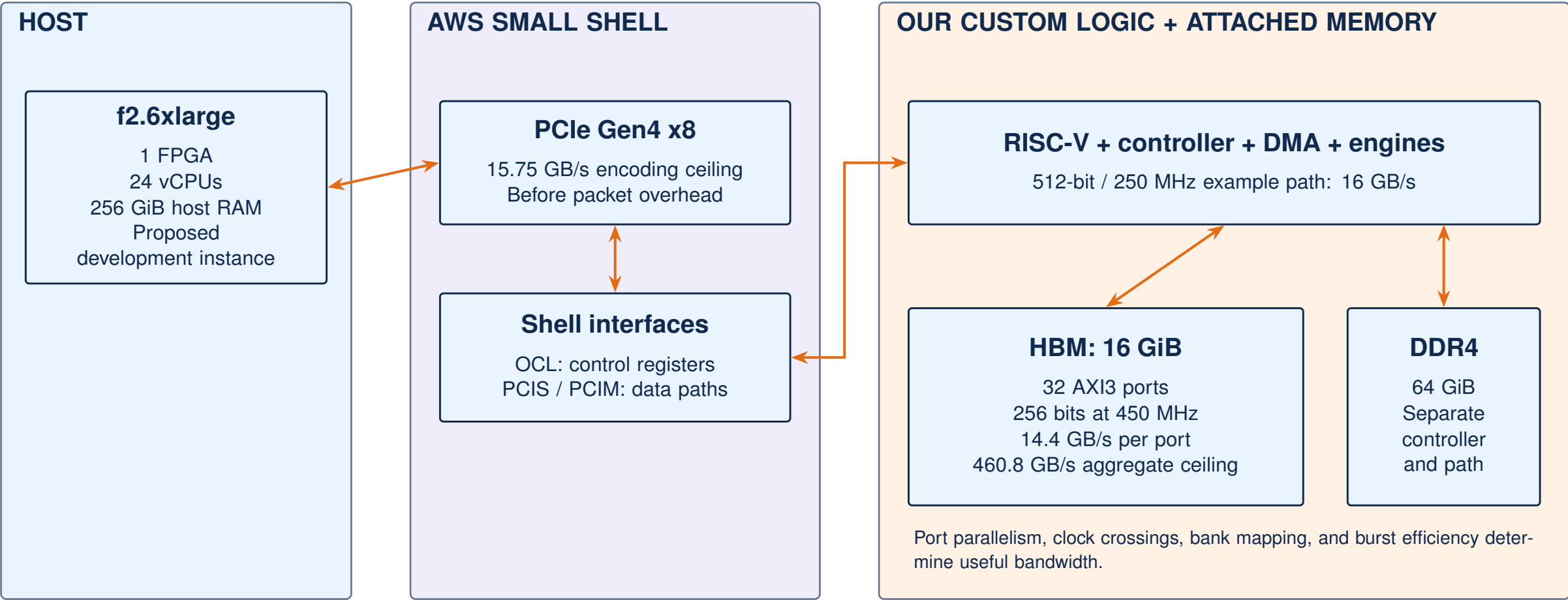


Takeaway. Specify who owns each buffer at every state. Never treat elapsed time as proof that DMA has stopped.

Sources and evidence: [6, 21, 26].

F2 has several bandwidth limits, not one

The host link, shell datapaths, HBM ports, and DDR controller are separate resources.

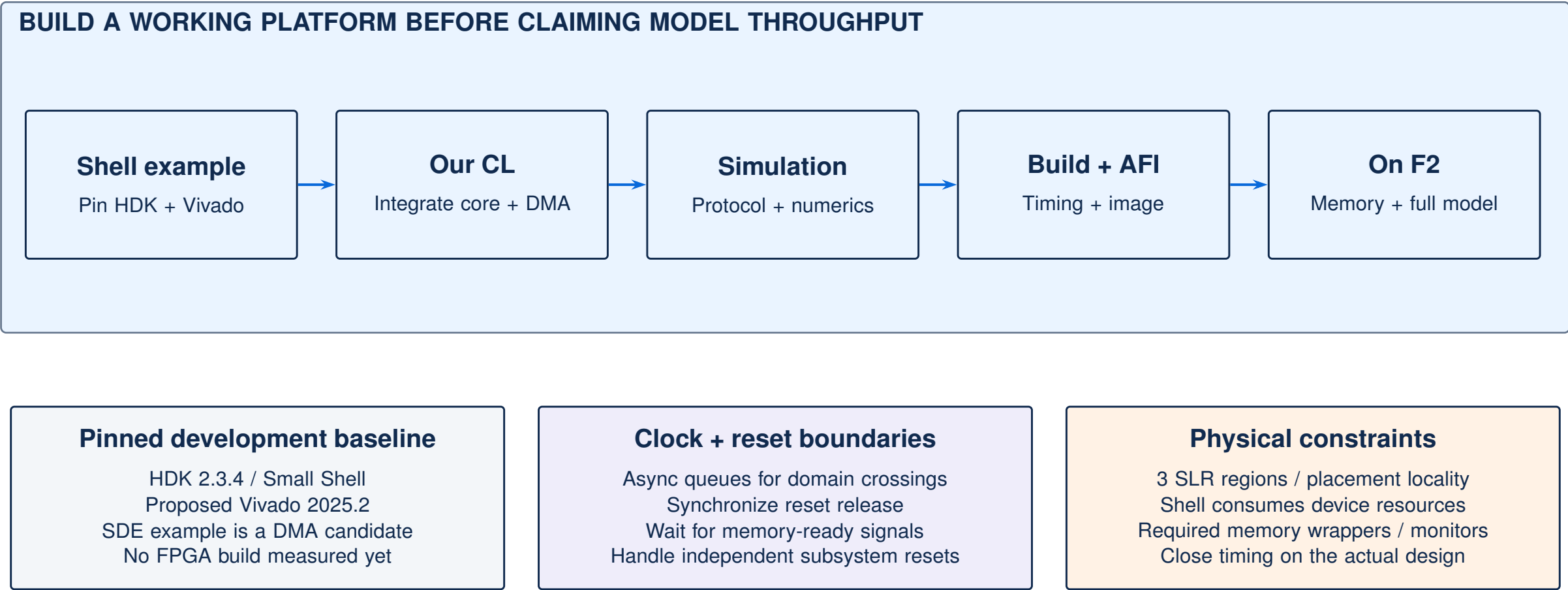


Takeaway. Reaching all HBM addresses through one port does not use all 32 ports in parallel. Measure the actual access pattern.

Sources and evidence: [1, 2, 20, 27, 28].

An FPGA port includes the shell and the physical design

Memory wrappers, clocks, reset ordering, and timing closure are part of the implementation.

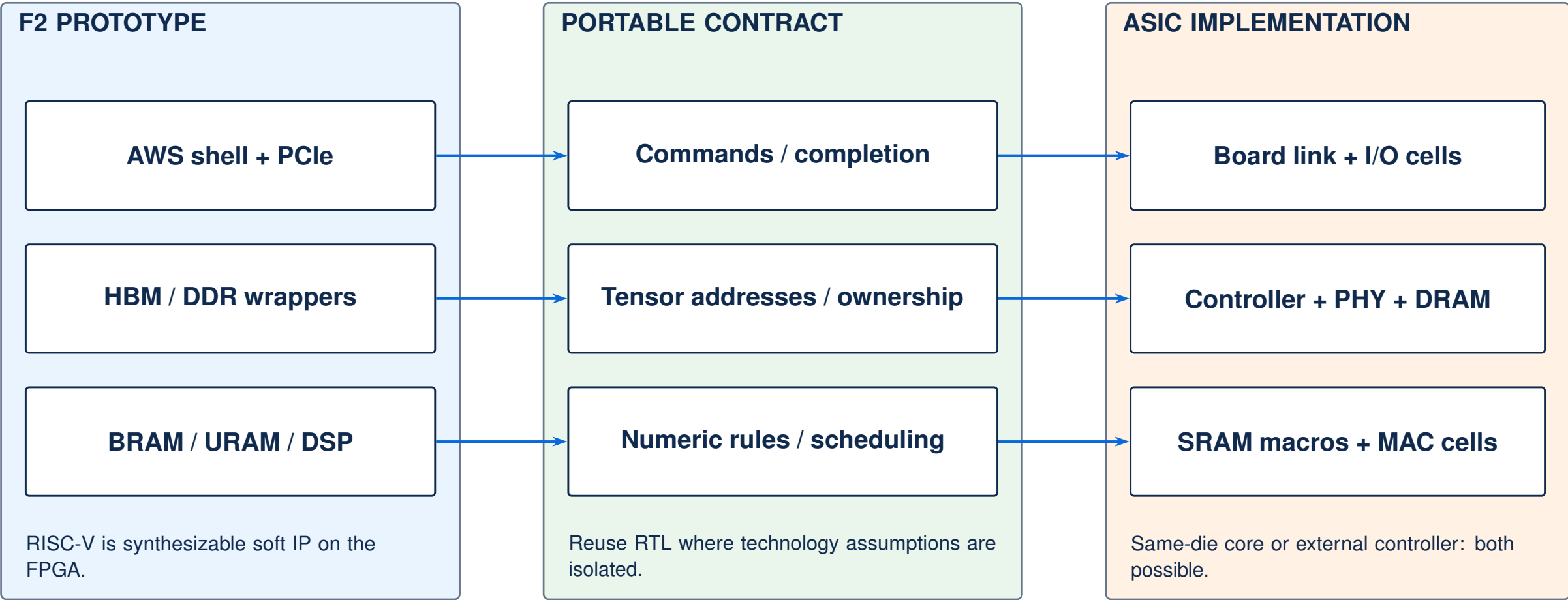


Takeaway. Do not infer achievable clock, usable SRAM, or model speed from the FPGA device headline resources.

Sources and evidence: [2, 29-33].

FPGA success does not make the design tapeout-ready

Keep numerical behavior and software interfaces stable while replacing platform-specific blocks.



Takeaway. RISC-V describes an ISA. A chosen core is an RTL IP block; SRAM, PHYs, package I/O, and sign-off are separate decisions.

Sources and evidence: [2, 23, 34-36].

A chip needs both enough DRAM and a fast enough link

A small external controller can manage commands, but the bulk tensor path determines performance.

Capacity check

About 1.002 GB of weights + 114.84 MiB of 8K request state
Before workspace, padding, and multi-request capacity

Bandwidth check

Counted traffic: 1.1407 GB/token at 8K
Illustrative 20 token/s point: 22.81 GB/s useful

4-bit SDR, 100 MHz

0.05 GB/s raw
About 22.8 s/token
Small demo interface

32-bit, 3200 MT/s

12.8 GB/s raw
Below 22.81 GB/s
Even at ideal efficiency

64-bit, 3200 MT/s

25.6 GB/s raw
Needs 89% useful
Little overhead margin

THESE ARE ARITHMETIC EXAMPLES, NOT QUALIFIED PROCESS INTERFACES

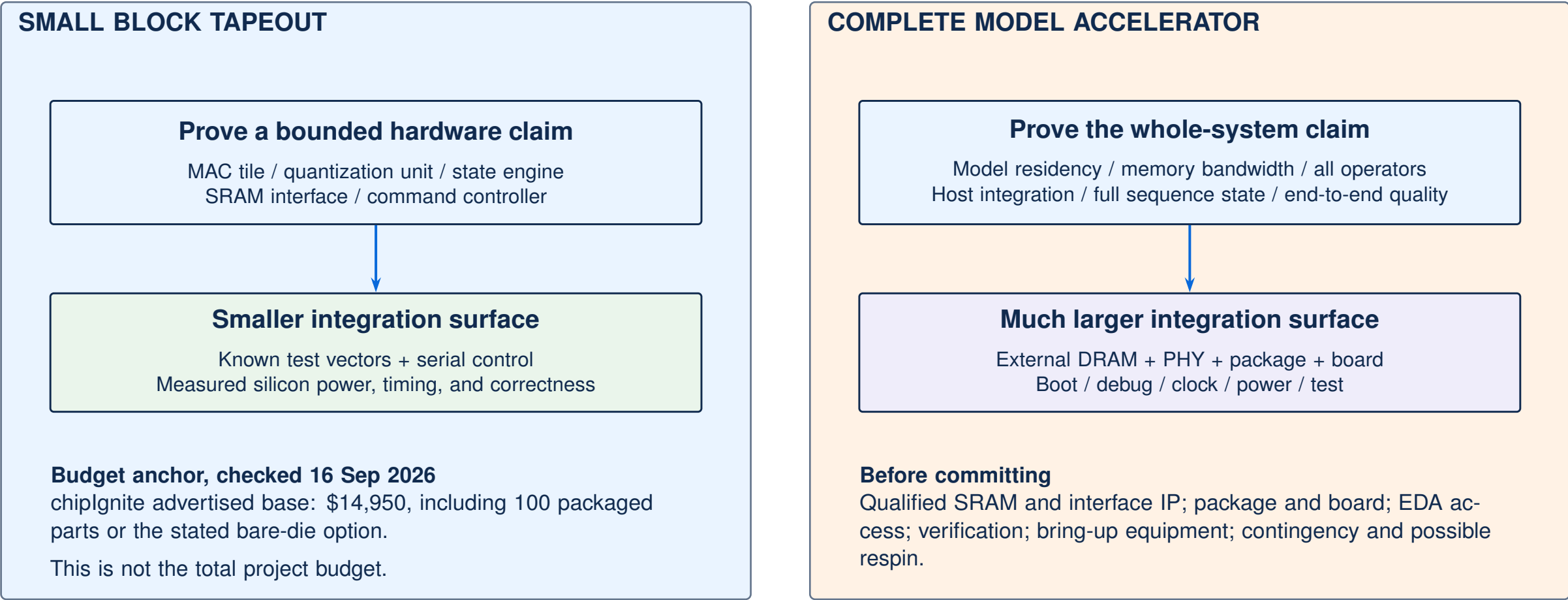
Every real choice needs a controller, PHY, timing budget, package pins, board routing, and verified memory parts. HBM also requires an appropriate package and very wide die-to-memory interface.

Takeaway. An external RISC-V processor does not need to share a package. It must not become the bulk-data bottleneck.

Sources and evidence: [7, 37, 38].

Choose the silicon claim before choosing the shuttle

A block-validation chip and an end-to-end Qwen accelerator are different scope and budget decisions.

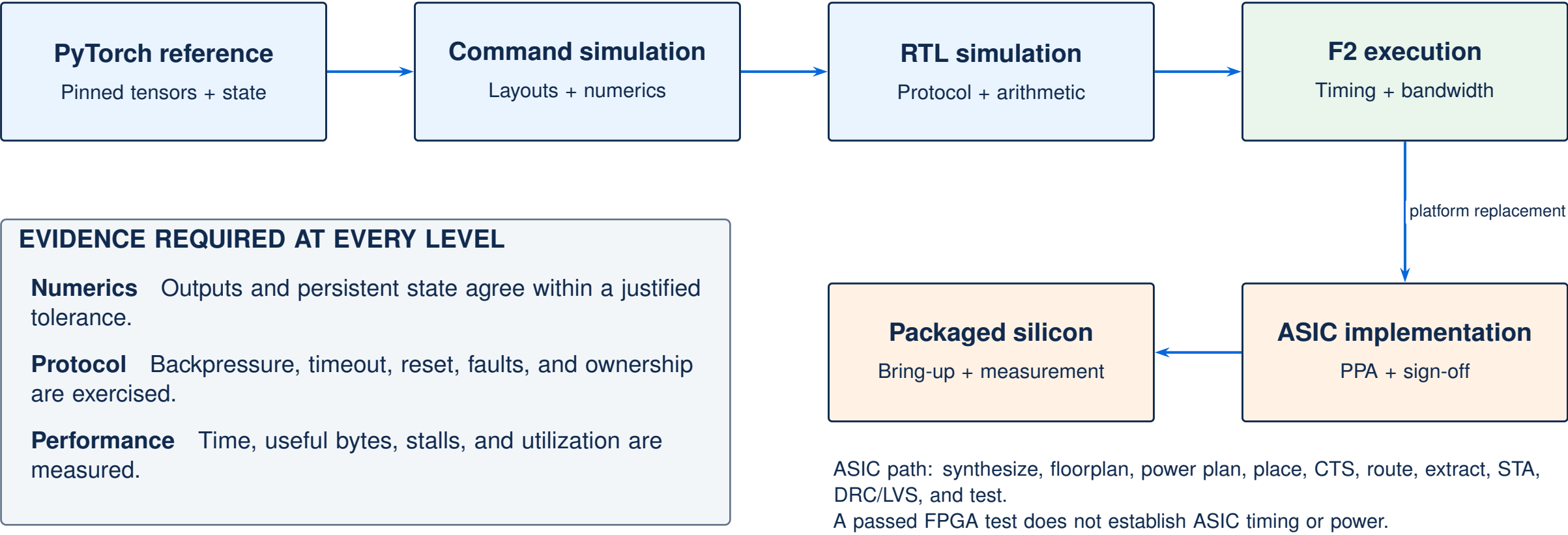


Takeaway. Keep FPGA full-model inference as an explicit milestone. A smaller chip can validate a component without replacing that goal.

Sources and evidence: [37-41].

Each milestone must leave reproducible evidence

Correctness comes from cross-checking tensors and state, not only watching plausible generated text.

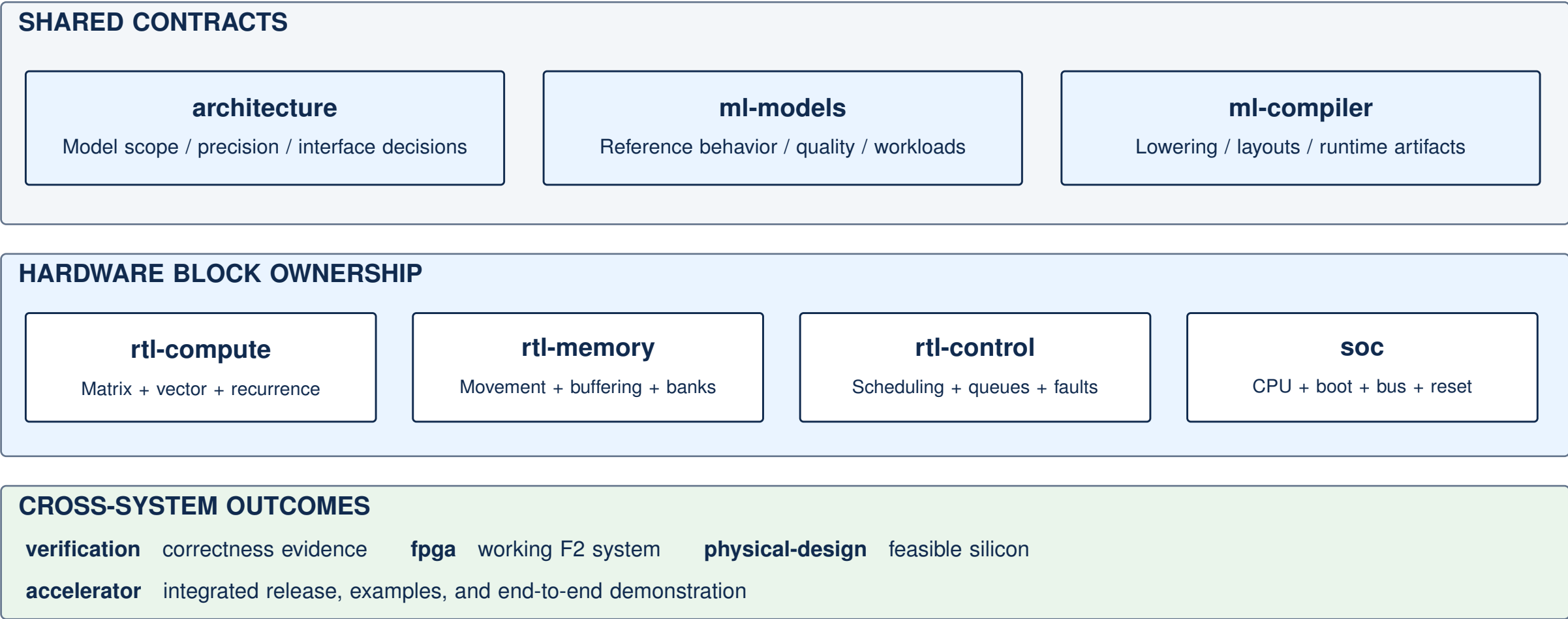


Takeaway. Advance on measured exit criteria. Keep software, FPGA, and silicon claims attached to the stage that produced them.

Sources and evidence: [8, 10, 35, 36, 41].

Split teams by technical responsibility and shared outcomes

Repositories are charters with interface obligations. Members choose how to achieve those outcomes.

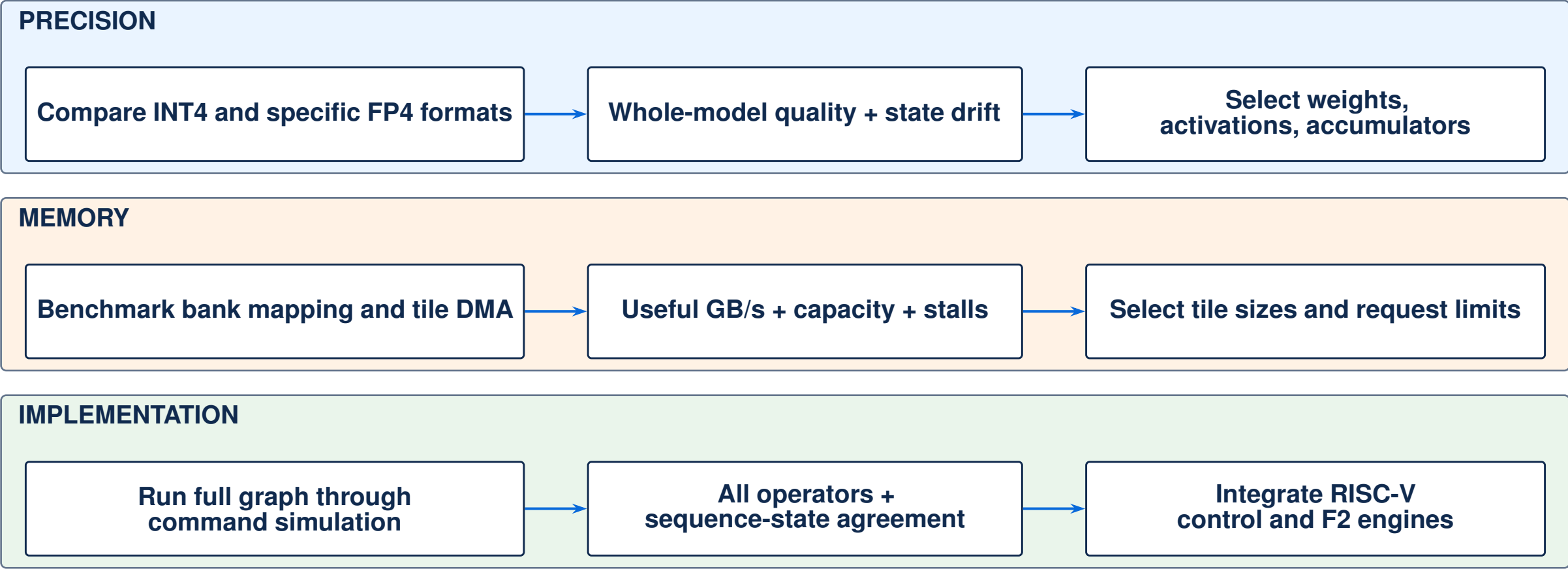


Takeaway. Control owns accelerator execution. SoC owns the CPU and system infrastructure. The boundary should stay explicit.

Project proposal: repository responsibilities, with no individual assignments.

Three decisions unlock the next architecture revision

The evidence already identifies the full graph and state. The next work resolves quality and hardware efficiency.

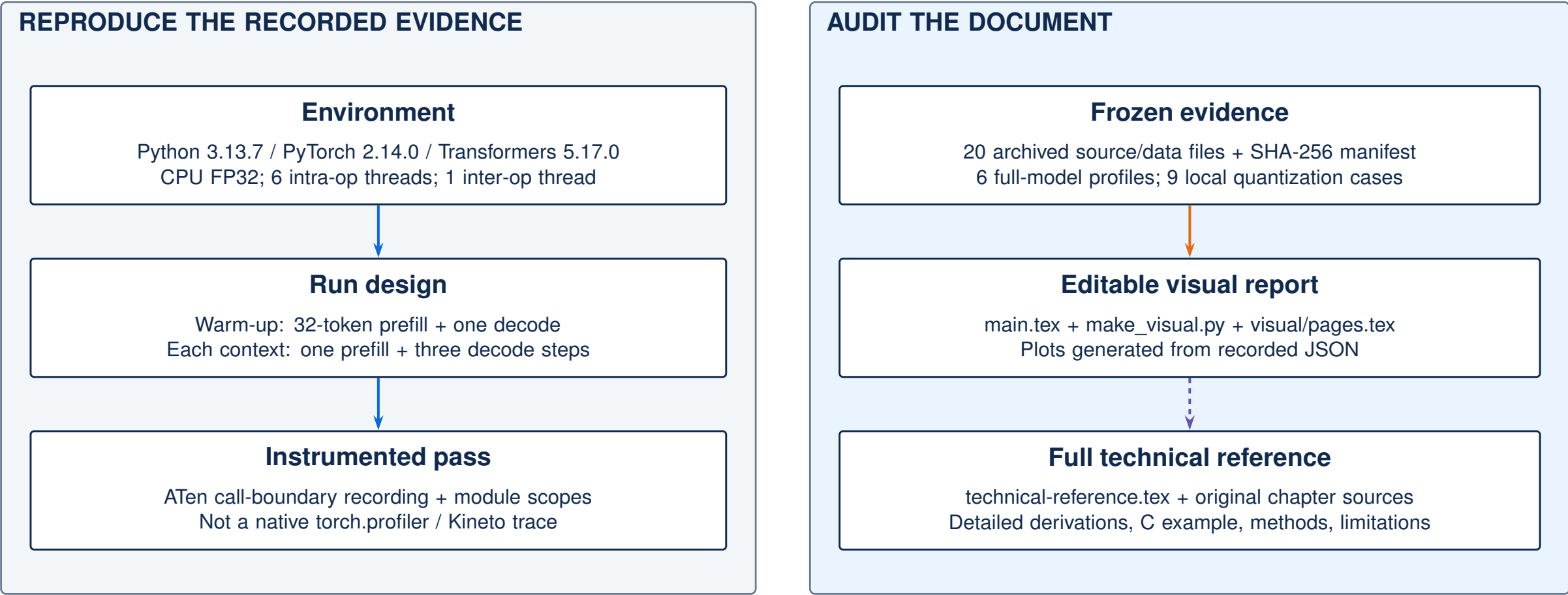


Takeaway. Freeze choices when their evidence is ready. Keep the model contract stable while implementations evolve.

Sources and evidence: [7-9].

Appendix: trace every figure back to its evidence

The source bundle preserves the detailed report while making this version easier to scan.



Takeaway. Measured profiles cover all 24 layers and all 187 linear calls. Hardware performance and quantized model quality remain future evidence.

Sources and evidence: [\[5, 8-10\]](#).

Sources behind the diagrams

Primary documentation, model implementation, research papers, and recorded project evidence.

[1] Amazon Web Services. *Amazon EC2 F2 Instances*. 2026. URL: <https://aws.amazon.com/ec2/instance-types/f2/> (visited on 09/16/2026).

[2] Amazon Web Services. *AWS Shell Interface Specification*. 2026. URL: https://github.com/aws/aws-fpga/blob/b603a81f65666e0cf7a67ee5cf18b148eb6b08c3/hdk/docs/AWS_Shell_Interface_Specification.md (visited on 09/16/2026).

[3] RISC-V International. *RV32I Base Integer Instruction Set, Version 2.1*. 2026. URL: <https://docs.riscv.org/reference/isa/v20260120/unpriv/rv32.html> (visited on 09/16/2026).

[4] ggml-org. *llama.cpp: ggml backend interface*. 2026. URL: <https://github.com/ggml-org/llama.cpp/blob/master/ggml/src/ggml-backend-impl.h> (visited on 09/16/2026).

[5] Qwen. *Qwen3.5-2B pinned checkpoint configuration*. 2026. URL: <https://huggingface.co/Qwen/Qwen3.5-2B/blob/15852e8c16360a2fea060d615a32b45270f8a8fc/config.json> (visited on 09/16/2026).

[6] Hugging Face Transformers. *Qwen3.5 modeling implementation*. 2026. URL: https://github.com/huggingface/transformers/blob/5b7dcb0d36c242d8d85920a81c564ef3a86ca6dd/src/transformers/models/qwen3_5/modeling_qwen3_5.py (visited on 09/16/2026).

[7] SiliconBadgers. *Model and F2 analytical budgets*. Bundled evidence/planning/model-budget.json and f2-budget.json. Derived values with explicit assumptions. 2026.

[8] SiliconBadgers. *Whole-model Qwen3.5-2B CPU profiling record*. Run on 16 September 2026 UTC. Bundled evidence/experiments/whole-model/results.json, metadata.json, and validation.json. 2026.

[9] SiliconBadgers. *Inference, isolated INT4 MLP quantization, and export experiments*. Run on 16 September 2026 UTC. Bundled evidence/experiments/ metadata, inference, quantization, export, and command JSON files. 2026.

[10] SiliconBadgers. *Original experiment and profiling implementations*. Bundled evidence/scripts/run_experiments.py and profile_whole_model.py. Fingerprints in evidence-manifest.json. 2026.

[11] PyTorch. *PyTorch 2.14 profiler documentation*. 2026. URL: <https://docs.pytorch.org/docs/2.14/profiler.html> (visited on 09/16/2026).

[12] Ji Lin et al. *AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration*. 2024. URL: <https://arxiv.org/abs/2306.00978> (visited on 09/16/2026).

[13] Elias Frantar et al. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers*. 2023. URL: <https://arxiv.org/abs/2210.17323> (visited on 09/16/2026).

[14] Open Compute Project. *Microscaling Formats (MX) Specification, Version 1.0*. 2023. URL: <https://www.opencompute.org/documents/ocp-microscaling-formats-mx-v1-0-spec-final-pdf> (visited on 09/16/2026).

[15] NVIDIA. *Introducing NVFP4 for Efficient and Accurate Low-Precision Inference*. 2025. URL: <https://developer.nvidia.com/blog/introducing-nvfp4-for-efficient-and-accurate-low-precision-inference/> (visited on 09/16/2026).

[16] Songlin Yang et al. *Gated Delta Networks: Improving Mamba2 with Delta Rule*. 2025. URL: <https://arxiv.org/abs/2412.06464> (visited on 09/16/2026).

[17] Joshua Ainslie et al. *GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. 2023. URL: <https://arxiv.org/abs/2305.13245> (visited on 09/16/2026).

[18] Tri Dao et al. *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*. 2022. URL: <https://arxiv.org/abs/2205.14135> (visited on 09/16/2026).

[19] PyTorch. *torch.export tutorial*. 2026. URL: https://docs.pytorch.org/tutorials/intermediate/torch_export_tutorial.html (visited on 09/16/2026).

[20] Amazon Web Services. *CL Memory Performance Example*. 2026. URL: https://github.com/aws/aws-fpga/blob/b603a81f65666e0cf7a67ee5cf18b148eb6b08c3/hdk/cl/examples/cl_mem_perf/README.md (visited on 09/16/2026).

[21] RISC-V International. *RVWMO Explanatory Material, Version 0.1*. 2026. URL: <https://docs.riscv.org/reference/isa/v20260120/unpriv/mm-eplan.html> (visited on 09/16/2026).

[22] RISC-V International. *Zicsr Extension for Control and Status Register Instructions, Version 2.0*. 2026. URL: <https://docs.riscv.org/reference/isa/v20260120/unpriv/zicsr.html> (visited on 09/16/2026).

Platform, implementation, and silicon sources

Source snapshots and the full technical reference are included in the LaTeX bundle.

[23]

lowRISC. *Ibex integration documentation*. 2026. URL: https://github.com/lowRISC/ibex/blob/master/doc/02_user/integration.rst (visited on 09/16/2026).

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[26]

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[28]

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[29]

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[31]

Amazon Web Services. *Amazon FPGA Images Guide*. 2026. URL: https://github.com/aws/aws-fpga/blob/b603a81f65666e0cf7a67ee5cf18b148eb6b08c3/hdk/docs/Amazon_FPGA_Images_Afis_Guide.md (visited on 09/16/2026).

[32]

Amazon Web Services. *AWS FPGA HDK errata*. 2026. URL: <https://github.com/aws/aws-fpga/blob/b603a81f65666e0cf7a67ee5cf18b148eb6b08c3/ERRATA.md> (visited on 09/16/2026).

[33]

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